LAB 7: FUZZ TESTING FOR INPUT VALIDATION BUGS

Blood Donation Screening Application

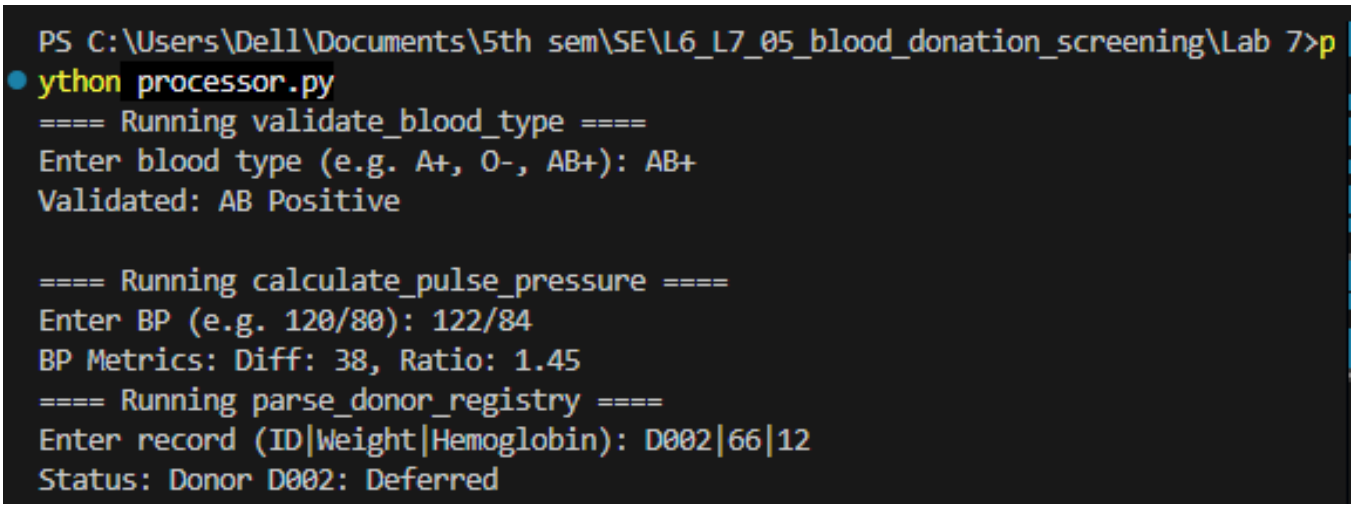
Screenshot Evidence Document

This document presents the execution evidence in sequence: original application, fuzz-test failures, fixed application, and final passing tests.

Final test result: 6 passed in 2.17 seconds

1. Original application execution

The original blood donation screening application was run with sample blood type, blood pressure, and donor registry inputs.



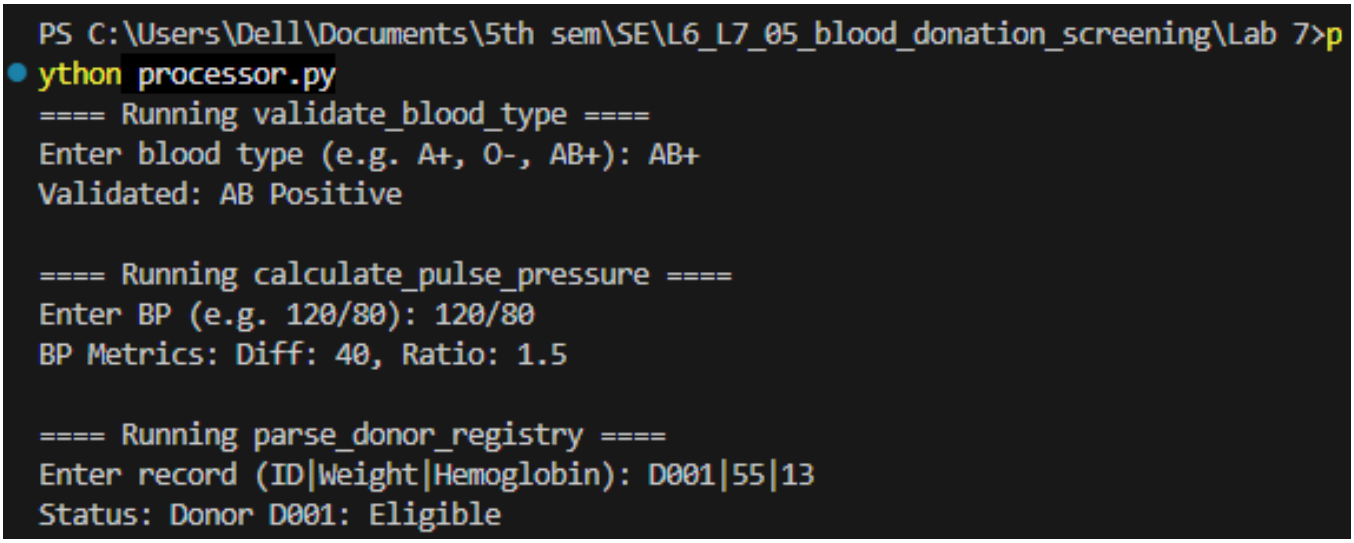
2. Hypothesis fuzz testing before fixing

Hypothesis exposed input-validation failures in all three functions. The run reported 3 failed and 3 passed tests.

```
PS C:\Users\Dell\Documents\5th sem\SE\L6_L7_05_blood_donation_screening\Lab 7>p
ython -m pytest test_processor.py -v
    |   File "C:\Users\Dell\Documents\5th sem\SE\L6_L7_05_blood_donation_screen
ing\Lab 7\test_processor.py", line 38, in test_parse_donor_registry_safe
    | AssertionError: parse_donor_registry crashed with None: 'NoneType' object
 has no attribute 'split'
    | assert False
    | Failing test case: test_parse_donor_registry_safe(
    |     value=None,  # or any other generated value
    | )
    +------------------------------------
========================== short test summary info ==========================
FAILED test_processor.py::test_validate_blood_type_no_crash - ExceptionGroup: H
ypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_processor.py::test_calculate_pulse_pressure_safe - ExceptionGroup:
Hypothesis found 3 distinct failures. (3 sub-exceptions)
thesis found 3 distinct failures. (3 sub-exceptions)
======================== 3 failed, 3 passed in 4.13s ========================
```

3. Fixed application execution

The corrected application processed the sample inputs and returned the screening results without crashing.



4. Final Hypothesis test results

After fixing the application, all 6 tests passed. The final run completed in 2.17 seconds.

